



Custom oscillator for KORG logue SDK 2.0 synthesizers

Operations Manual

v.3.0-9

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Introduction

FM64 is a custom oscillator for KORG NTS-1 mkII and NTS-3 synthesizers that reproduces Yamaha DX / TX series 6-operator FM synthesis with several additional features from Yamaha SY77 series and KORG opsix.

The oscillator must be pre populated with the Yamaha voice banks of your choice using the online constructor (see [Quick Start](#) section) before uploading to the synthesizer. For information on how to upload a custom oscillator to the synthesizer and how to activate it, please refer to the Synthesizer Owner's Manual and Sound Librarian Owner's Manual for your KORG synthesizer model.

Yamaha voice banks, or ROMs, are widely available on the Internet. Any VMEM packed voice bank for Yamaha DX1, DX5, DX7, DX7II, DX7s, TX7, TX802, TX816 both in SysEx (4104 bytes) or RAW (4096 bytes) will fit. Any other format, including voice banks for 4-operator Yamaha DX9, DX11, DX21, DX27s, DX100, TX81Z, will not work.

All the oscillator customization operations performed by the online constructor are done in JavaScript of your browser, so no actual upload occurs. Online constructor does not store any data, except for the your browser cookie setting for the last selected synthesizer model.

If you find a bug or wish to propose a new feature or improvement, don't hesitate to create a new issue at [GitHub](#) or just send me an email to dukesrg@gmail.com.

This custom oscillator is open source and free. However you can support the development via [PayPal me](#) or [Revolut me](#).

Quick start

The raw oscillator file has no banks inside and won't produce any sound. To make the oscillator work you must first populate it with the voice banks.

Obtaining the oscillator with custom voices

1. Navigate to the [online constructor](#) web page.
2. Select your KORG synthesizer model to define the target format of the oscillator file.
3. Locate the FM64 oscillator row by the column **NAME**
4. Check the **SIZE** column of this row, the last multiplier is the maximum number of voice banks this oscillator can contain.
5. Click the **Upload** button located in the **CUSTOM DATA** column of this row.
6. In the file open dialog select one to several (up to obtained in step 4) voice bank files.
7. Check the **CUSTOM NAME** cell in this row. This name is generated from the names of the uploaded banks and you can alter it now. This name will be displayed by the Librarian and your synthesizer.
8. Click the **Download** button located in the **CUSTOM UNIT** cell of this row.
9. Now you can upload the oscillator file to your KORG synthesizer with the Librarian application.

The screenshot shows the 'prologue' online constructor interface. At the top, there are tabs for 'prologue', 'minilogue xd', and 'NTS-1 digital'. A red circle with the number '2' is placed over the 'NTS-1 digital' tab. Below the tabs is a table of oscillators. The table has columns: NAME, VERSION, API, SHAPE, ALT, PARAM 1, PARAM 2, PARAM 3, PARAM 4, PARAM 5, PARAM 6, CUSTOM DATA, CUSTOM NAME, CUSTOM UNIT, SIZE, and DESCRIPTION. The row for 'FM64' is highlighted. Red circles with numbers 5, 7, 8, and 4 are placed over the 'CUSTOM DATA', 'CUSTOM NAME', 'CUSTOM UNIT', and 'SIZE' columns of the 'FM64' row, respectively. The 'CUSTOM DATA' column contains an 'Upload' button. The 'CUSTOM NAME' column contains a text input field. The 'CUSTOM UNIT' column contains a 'Download' button. The 'SIZE' column shows '4096 x 4'.

NAME	VERSION	API	SHAPE	ALT	PARAM 1	PARAM 2	PARAM 3	PARAM 4	PARAM 5	PARAM 6	CUSTOM DATA	CUSTOM NAME	CUSTOM UNIT	SIZE	DESCRIPTION
1. FM63	1.5.0	1.5.0	Velocity	FB scale	Voice 1	Bank	Shape Assign	All Assign	S LFO Target	Algorithm	Upload		Download	4096 x 4	Yamaha DX2/DX2T1 series voice banks
2. FM64	1.7.3	1.5.0	Velocity	FB scale	Voice 1	Det scal All	Rel off All	Lvl off Car	Algorithm	Lvl off Mod	Upload		Download	4096 x 5	Yamaha DX7 series voice banks
3. FM65	1.7.3	1.5.0	Velocity	FB scale	Voice 1	Det scal All	Rel off All	Lvl off Car	Algorithm	Waveform C-M	Upload		Download	4096 x 4 512 x 8	Yamaha DX7 series voice banks Pace 10-M PCM waveforms
4. FM67	1.7.3	1.5.0	Velocity	FB scale	Voice 1	Det scal All	Rel off All	Lvl off Car	Algorithm	Waveform C-M	Upload		Download	4096 x 3 512 x 16	Yamaha DX7 series voice banks Pace 10-M PCM waveforms
5. FM68	1.5.0	1.5.0	Velocity	FB scale	Voice 1	Bank	Shape Assign	All Assign	S LFO Target	Algorithm	Upload		Download	4096 x 2 1024 x 8	Yamaha DX7 series voice banks Pace 10-M PCM waveforms
6. FM69	1.7.3	1.5.0	Velocity	FB scale	Transpose 1	Voice Shift1	Split Point1	Voice Shift2	Split Point2	Voice Shift3	Upload		Download	4096 x 5	Yamaha DX7 series voice banks
7. Morphous	1.0-1	1.5.0	Velocity	FB scale	Mode	LFOX	LFOY	LFO Trigger	Interpolate		Upload		Download	16384 x 1	Pace wave p-lar encoded
8. Anthology	0.7.6	1.5.0	Velocity	FB scale	Prog	Sub	Mode	AC 1	AC 2		Upload		Download	1024 x 25	Korg legacy series program data libraries
9. —	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
10. —	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
11. —	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
12. —	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
13. —	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
14. —	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
15. —	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
16. —	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—

Changing oscillator custom parameters

1. Proceed with steps 1 thru 7 of the [previous section](#).
2. Click on one of the highlighted values in the columns **SHAPE/A**, **ALT/B**, **PARAM 1**, **PARAM 2**, **PARAM 3**, **PARAM 4**, **PARAM 5**, **PARAM 6**, **PARAM 7**, **PARAM 8** of this row.
3. From the popup menu select the desired custom parameter for the parameter selected in step 2. You need to scroll with the mouse wheel to reach all of the available custom parameters.
4. Repeat steps 2 and 3 for other oscillator parameters you wish to reassign.
5. Proceed with steps 8 and 9 of the [previous section](#).

The screenshot shows the Prologue software interface with the 'minilogue xd' and 'NTS-1 digital' tabs selected. A table of oscillator parameters is displayed. The table has columns for NAME, VERSION, API, SHAPE, ALT, PARAM 1, PARAM 2, PARAM 3, PARAM 4, PARAM 5, PARAM 6, CUSTOM DATA, CUSTOM NAME, CUSTOM UNIT, SIZE, and DESCRIPTION. Row 2 is highlighted, and a red box is drawn around the 'PARAM 3' column for this row. A red circle with the number '2' is placed above the 'PARAM 3' column. Another red box is drawn around the 'Velocity' parameter in the 'PARAM 3' column for row 2, and a red circle with the number '3' is placed next to it.

NAME	VERSION	API	SHAPE	ALT	PARAM 1	PARAM 2	PARAM 3	PARAM 4	PARAM 5	PARAM 6	CUSTOM DATA	CUSTOM NAME	CUSTOM UNIT	SIZE	DESCRIPTION	
1	FM4	1.5.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
2	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
3	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
4	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
5	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
6	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
7	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
8	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
9	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
10	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
11	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
12	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
13	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
14	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
15	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks
16	FM4	1.7.0	1.5.0	Velocity	FB scale	Voice 1	Oct scal All	Rel offs All	Lvl offs Car	Algorithm	Lvl offs Mod	Upload		Download	4096 x 4	Yamaha DX210/211 series voice banks

Advanced features

Due to the hardware differences of KORG NTS-1 mkII and NTS-3 synthesizers, several features are exclusive to each model.

NTS-1 mkII features

- Maximum voice banks is 4
- Channel Pressure and Aftertouch supported
- TBD

NTS-3 features

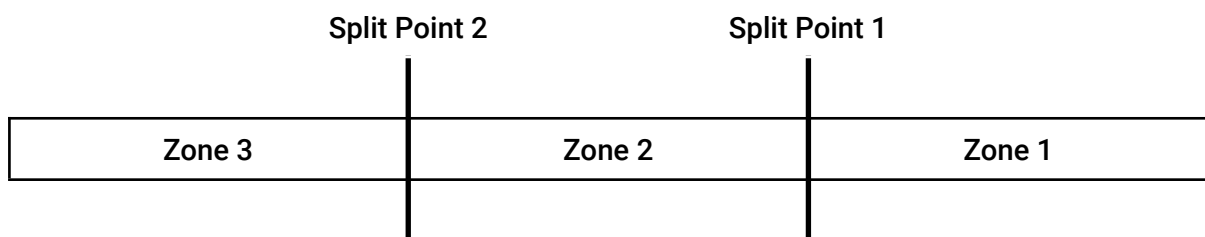
- Maximum voice banks is 1
- TBD

Velocity

Velocity can be controlled in different ways. By default the Note On event velocity is used. Using the Velocity custom parameter it is possible to override the velocity value with the enhanced 10-bit precision. The maximum Velocity parameter value of 127.75 enables Note On event velocity processing. While other values above 127.0 are saturated to that value. Channel Pressure and Aftertouch messages value also directly affects the velocity value in real time.

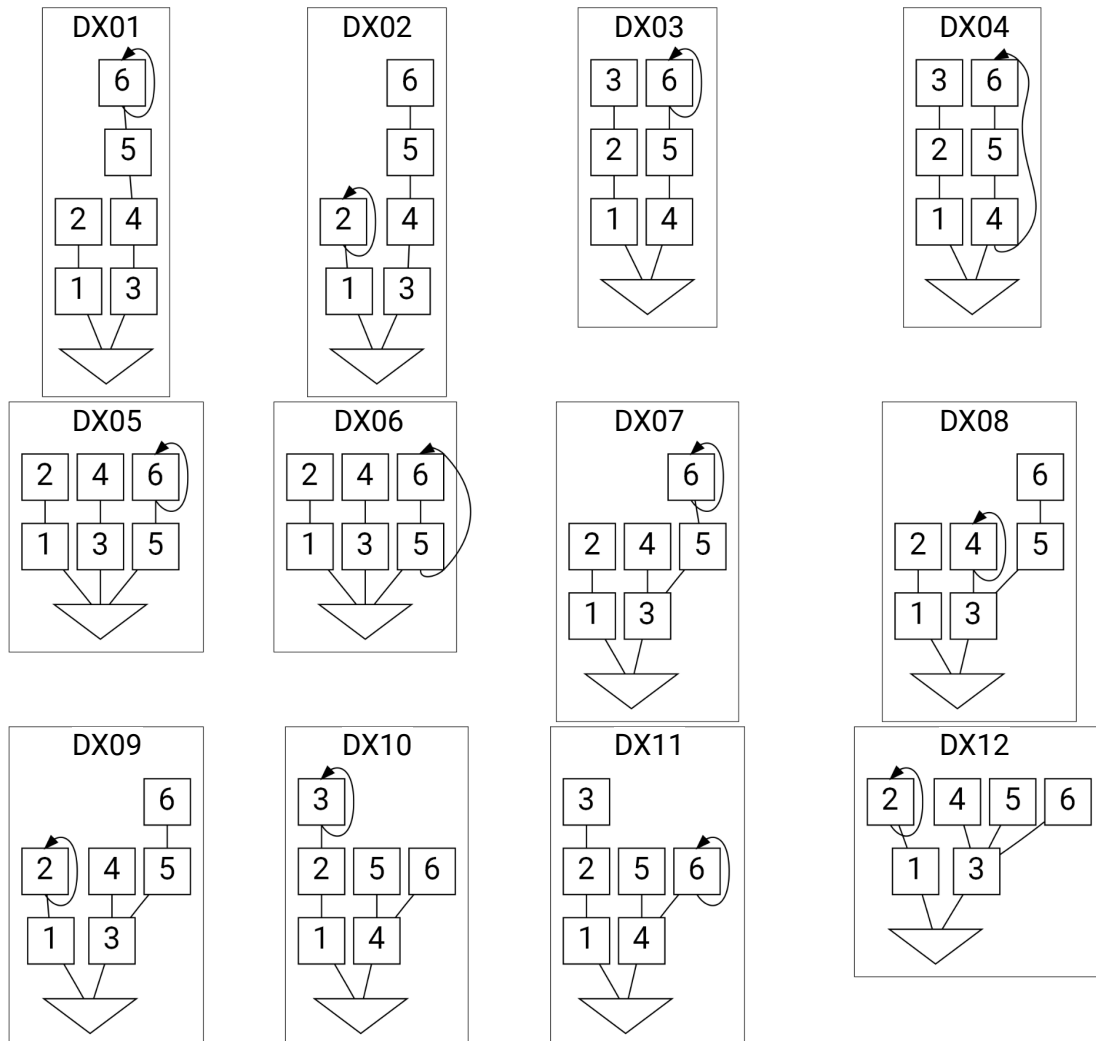
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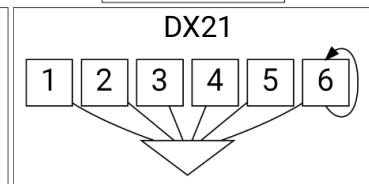
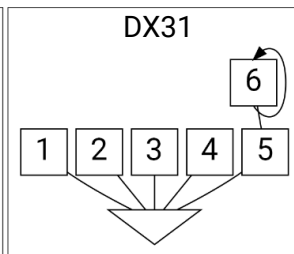
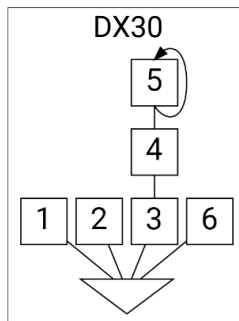
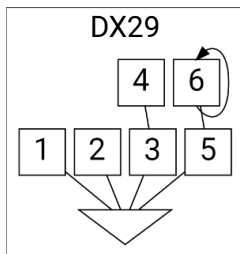
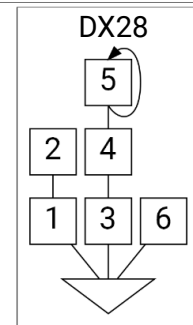
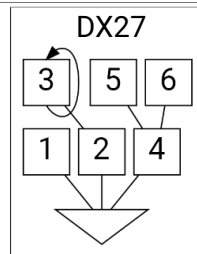
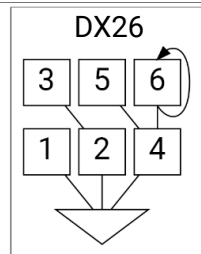
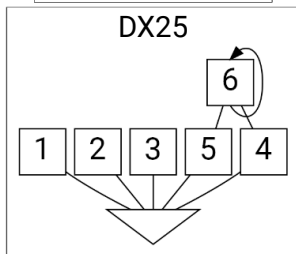
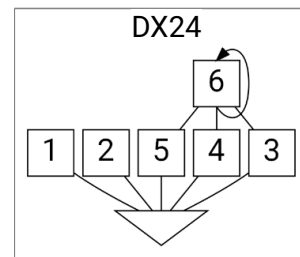
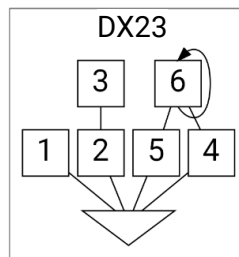
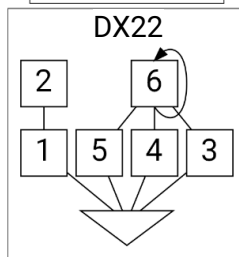
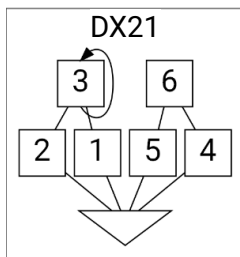
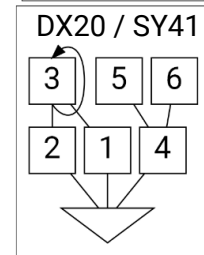
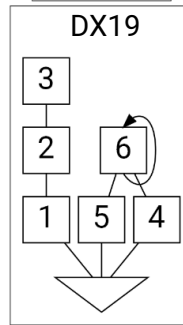
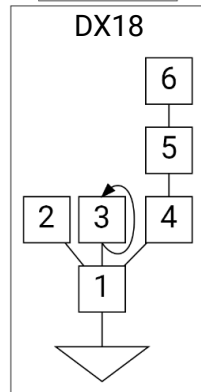
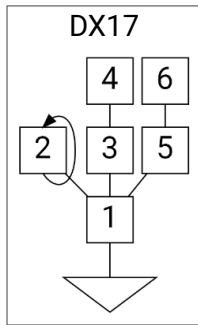
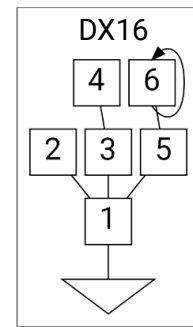
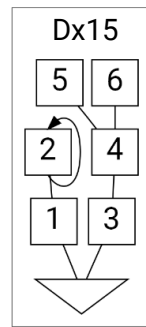
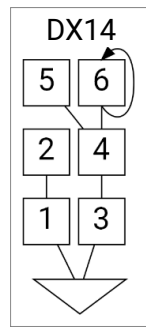
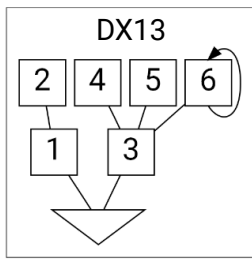
It is possible to split the keyboard to up to three zones and assign different voices to each of them. Split points determines the edge notes between two neighbor zones. Relative position of zones and split points are shown below:

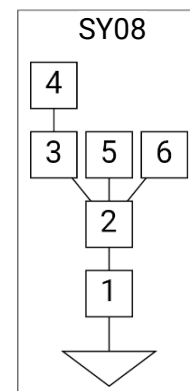
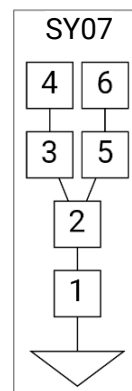
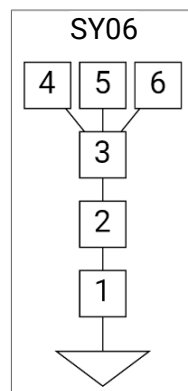
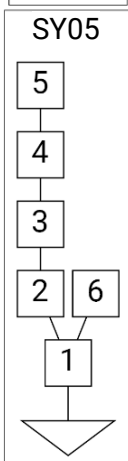
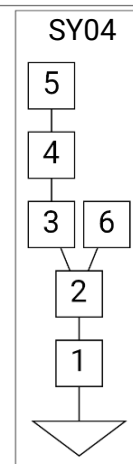
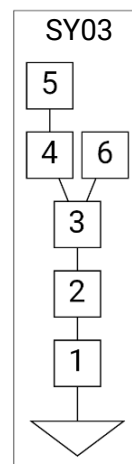
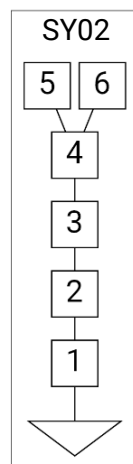
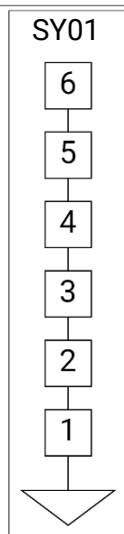
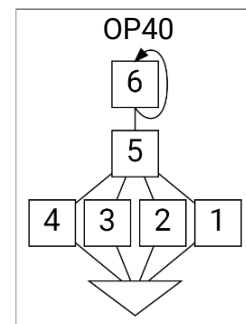
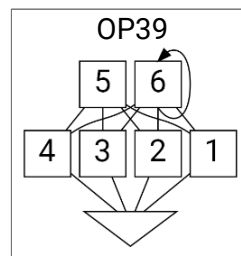
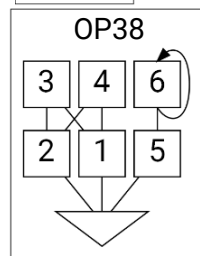
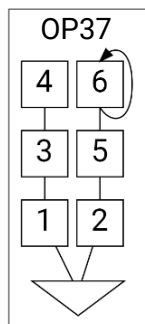
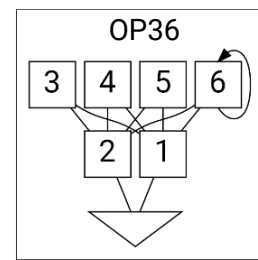
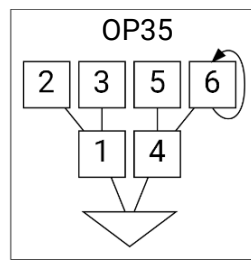
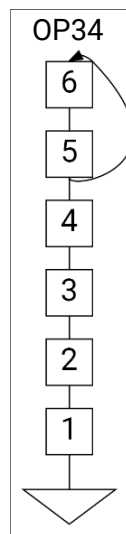
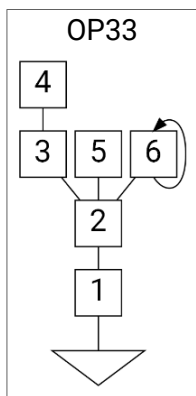


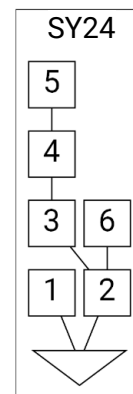
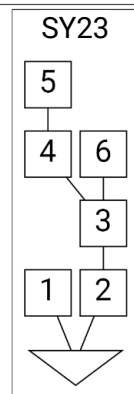
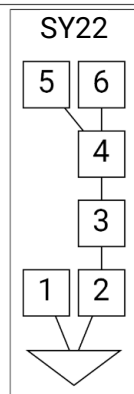
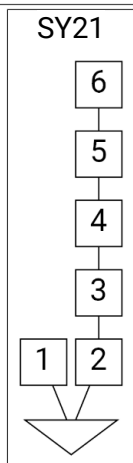
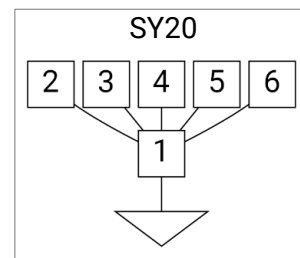
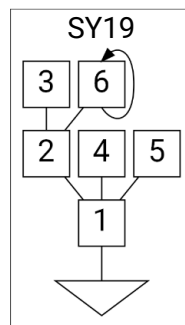
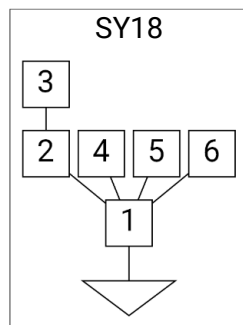
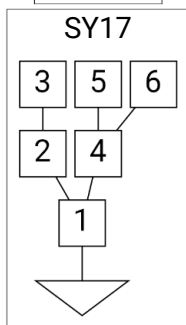
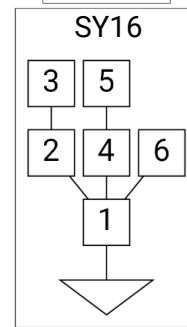
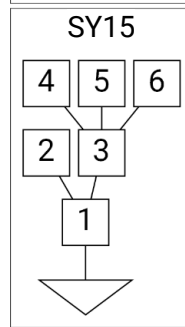
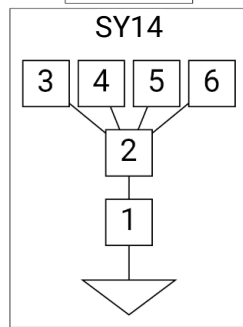
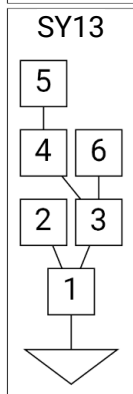
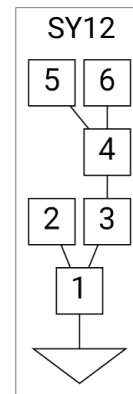
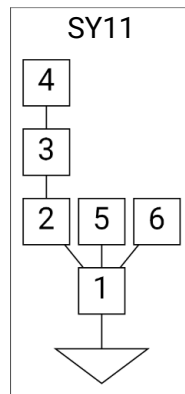
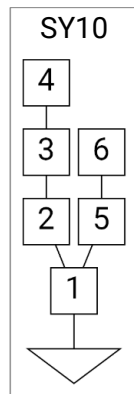
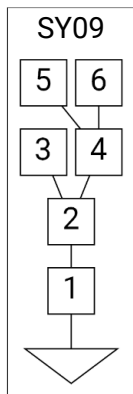
Algorithms list

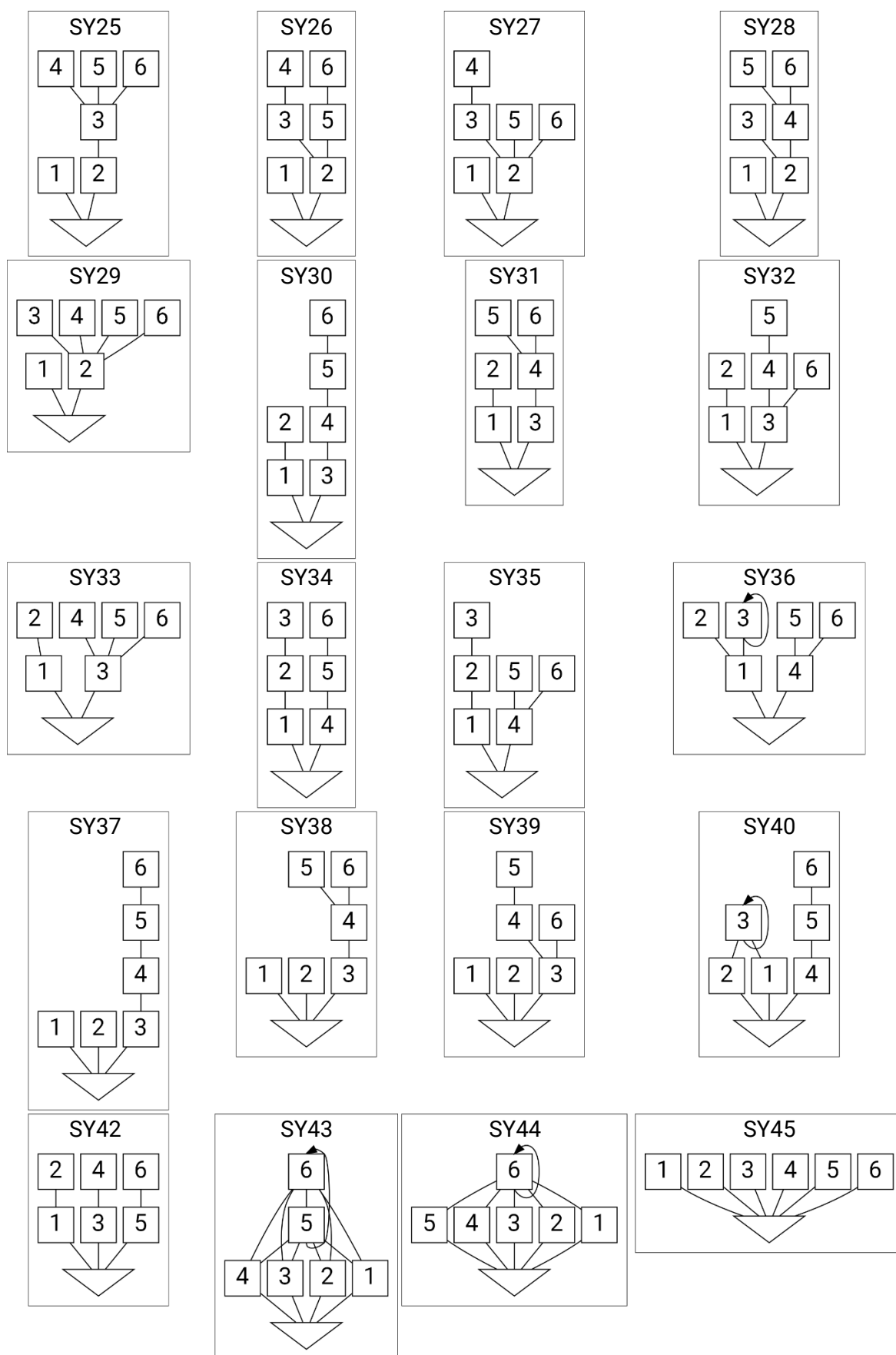
All oscillator variations support 32 Yamaha DX series and 8 additional KORG opsix algorithms. There are also 45 Yamaha SY77 / TG77 / SY99 algorithms supported with feedback count limitation, e.g. algorithm SY43 have a single feedback, unlike the original Yamaha SY77 series algorithm.











Custom parameters list

#	Custom param	Range	Description
0	Velocity	0.0...127.75 ⁽¹⁾⁽⁵⁾	Note velocity
1	Voice 1	0...127	Voice for zone 1
2	Voice 2	0...127	Voice for zone 2
3	Voice 3	0...127	Voice for zone 3
4	Split Point1	C-1...G9	Split point between zone 1 and 2
5	Split Point2	C-1...G9	Split point between zone 2 and 3
6	Transpose 1	-128...127 ⁽²⁾	Transpose for zone 1
7	Transpose 2	-128...127 ⁽²⁾	Transpose for zone 2
8	Transpose 3	-128...127 ⁽²⁾	Transpose for zone 3
9	Voice Shift1	-128...127	Voice shift for zone 1
10	Voice Shift2	-128...127	Voice shift for zone 2
11	Voice Shift3	-128...127	Voice shift for zone 3
12	-		
13	-		
14	FB offset	-8.0...7.984 ⁽⁵⁾	Feedback 1 offset
15	FB2 offset	-8.0...7.984 ⁽⁵⁾	Feedback 2 offset
16	FB scale	0.0...1.998 ⁽⁵⁾	Feedback 1 multiplier
17	FB2 scale	0.0...1.998 ⁽⁵⁾	Feedback 2 multiplier
18	FB route	0...36 ⁽³⁾⁽⁵⁾	Feedback 1 route
19	FB2 route	0...36 ⁽³⁾⁽⁵⁾	Feedback 2 route
20	Alg select	0...85 ⁽⁴⁾⁽⁵⁾	Algorithm select
21	Alg offset	-128...127 ⁽⁵⁾	Algorithm offset
22	Lvl offs All	-128.0...127.75	Level offset for all operators
23	Lvl offs Car	-128.0...127.75	Level offset for carriers
24	Lvl offs Mod	-128.0...127.75	Level offset for modulators
25	Lvl offs Op1	-128.0...127.75	Level offset for operator 1
26	Lvl offs Op2	-128.0...127.75	Level offset for operator 2
27	Lvl offs Op3	-128.0...127.75	Level offset for operator 3
28	Lvl offs Op4	-128.0...127.75	Level offset for operator 4
29	Lvl offs Op5	-128.0...127.75	Level offset for operator 5
30	Lvl offs Op6	-128.0...127.75	Level offset for operator 6

31	Lvl scal All	0.0...1.998	Level multiplier for all operators
32	Lvl scal Car	0.0...1.998	Level multiplier for carriers
33	Lvl scal Mod	0.0...1.998	Level multiplier for modulators
34	Lvl scal Op1	0.0...1.998	Level multiplier for operator 1
35	Lvl scal Op2	0.0...1.998	Level multiplier for operator 2
36	Lvl scal Op3	0.0...1.998	Level multiplier for operator 3
37	Lvl scal Op4	0.0...1.998	Level multiplier for operator 4
38	Lvl scal Op5	0.0...1.998	Level multiplier for operator 5
39	Lvl scal Op6	0.0...1.998	Level multiplier for operator 6
40	KLS offs All	-128.0...127.75	Keyboard level scaling offset for all operators
41	KLS offset Car	-128.0...127.75	Keyboard level scaling offset for carriers
42	KLS offset Mod	-128.0...127.75	Keyboard level scaling offset for modulators
43	KLS offset Op1	-128.0...127.75	Keyboard level scaling offset for operators 1
44	KLS offset Op2	-128.0...127.75	Keyboard level scaling offset for operators 2
45	KLS offset Op3	-128.0...127.75	Keyboard level scaling offset for operators 3
46	KLS offset Op4	-128.0...127.75	Keyboard level scaling offset for operators 4
47	KLS offset Op5	-128.0...127.75	Keyboard level scaling offset for operators 5
48	KLS offset Op6	-128.0...127.75	Keyboard level scaling offset for operators 6
49	KLS scal All	0.0...1.998	Keyboard level scaling multiplier for all operators
50	KLS scal Car	0.0...1.998	Keyboard level scaling multiplier carriers
51	KLS scal Mod	0.0...1.998	Keyboard level scaling multiplier modulators
52	KLS scal Op1	0.0...1.998	Keyboard level scaling multiplier for operator 1
53	KLS scal Op2	0.0...1.998	Keyboard level scaling multiplier for operator 2
54	KLS scal Op3	0.0...1.998	Keyboard level scaling multiplier for operator 3
55	KLS scal Op4	0.0...1.998	Keyboard level scaling multiplier for operator 4
56	KLS scal Op5	0.0...1.998	Keyboard level scaling multiplier for operator 5
57	KLS scal Op6	0.0...1.998	Keyboard level scaling multiplier for operator 6
58	KVS offs All	-8.0...7.984	Key velocity sensitivity offset for all operators
59	KVS offs Car	-8.0...7.984	Key velocity sensitivity offset for carriers
60	KVS offs Mod	-8.0...7.984	Key velocity sensitivity offset for operators
61	KVS offs Op1	-8.0...7.984	Key velocity sensitivity offset operator 1
62	KVS offs Op2	-8.0...7.984	Key velocity sensitivity offset operator 2
63	KVS offs Op3	-8.0...7.984	Key velocity sensitivity offset operator 3
64	KVS offs Op4	-8.0...7.984	Key velocity sensitivity offset operator 4

65	KVS offs Op5	-8.0...7.984	Key velocity sensitivity offset operator 5
66	KVS offs Op6	-8.0...7.984	Key velocity sensitivity offset operator 6
67	KVS scal All	0.0...1.998	Key velocity sensitivity multiplier for all operators
68	KVS scal Car	0.0...1.998	Key velocity sensitivity multiplier for carriers
69	KVS scal Mod	0.0...1.998	Key velocity sensitivity multiplier for modulators
70	KVS scal Op1	0.0...1.998	Key velocity sensitivity multiplier for operator 1
71	KVS scal Op2	0.0...1.998	Key velocity sensitivity multiplier for operator 2
72	KVS scal Op3	0.0...1.998	Key velocity sensitivity multiplier for operator 3
73	KVS scal Op4	0.0...1.998	Key velocity sensitivity multiplier for operator 4
74	KVS scal Op5	0.0...1.998	Key velocity sensitivity multiplier for operator 5
75	KVS scal Op6	0.0...1.998	Key velocity sensitivity multiplier for operator 6
76	Rat offs All	-128.0...127.75	EG rate offset for all operators
77	Rat offs Car	-128.0...127.75	EG rate offset for carriers
78	Rat offs Mod	-128.0...127.75	EG rate offset for modulators
79	Rat offs Op1	-128.0...127.75	EG rate offset for operator 1
80	Rat offs Op2	-128.0...127.75	EG rate offset for operator 2
81	Rat offs Op3	-128.0...127.75	EG rate offset for operator 3
82	Rat offs Op4	-128.0...127.75	EG rate offset for operator 4
83	Rat offs Op5	-128.0...127.75	EG rate offset for operator 5
84	Rat offs Op6	-128.0...127.75	EG rate offset for operator 6
85	Rat scal All	0.0...1.998	EG rate multiplier for all operators
86	Rat scal Car	0.0...1.998	EG rate multiplier for carriers
87	Rat scal Mod	0.0...1.998	EG rate multiplier for modulators
88	Rat scal Op1	0.0...1.998	EG rate multiplier for operator 1
89	Rat scal Op2	0.0...1.998	EG rate multiplier for operator 2
90	Rat scal Op3	0.0...1.998	EG rate multiplier for operator 3
91	Rat scal Op4	0.0...1.998	EG rate multiplier for operator 4
92	Rat scal Op5	0.0...1.998	EG rate multiplier for operator 5
93	Rat scal Op6	0.0...1.998	EG rate multiplier for operator 6
94	KRS offs All	-8.0...7.984	Keyboard EG rate scaling offset for all operators
95	KRS offs Car	-8.0...7.984	Keyboard EG rate scaling offset for carriers
96	KRS offs Mod	-8.0...7.984	Keyboard EG rate scaling offset for modulators
97	KRS offs Op1	-8.0...7.984	Keyboard EG rate scaling offset for operator 1
98	KRS offs Op2	-8.0...7.984	Keyboard EG rate scaling offset for operator 2

99	KRS offs Op3	-8.0...7.984	Keyboard EG rate scaling offset for operator 3
100	KRS offs Op4	-8.0...7.984	Keyboard EG rate scaling offset for operator 4
101	KRS offs Op5	-8.0...7.984	Keyboard EG rate scaling offset for operator 5
102	KRS offs Op6	-8.0...7.984	Keyboard EG rate scaling offset for operator 6
103	KRS scal All	0.0...1.998	Keyboard EG rate multiplier for all operators
104	KRS scal Car	0.0...1.998	Keyboard EG rate multiplier for carriers
105	KRS scal Mod	0.0...1.998	Keyboard EG rate multiplier for modulators
106	KRS scal Op1	0.0...1.998	Keyboard EG rate multiplier for operator 1
107	KRS scal Op2	0.0...1.998	Keyboard EG rate multiplier for operator 2
108	KRS scal Op3	0.0...1.998	Keyboard EG rate multiplier for operator 3
109	KRS scal Op4	0.0...1.998	Keyboard EG rate multiplier for operator 4
110	KRS scal Op5	0.0...1.998	Keyboard EG rate multiplier for operator 5
111	KRS scal Op6	0.0...1.998	Keyboard EG rate multiplier for operator 6
112	Det offs All	-128.0...127.75 ⁽⁵⁾	Detune offset in cents for all operators
113	Det offs Car	-128.0...127.75 ⁽⁵⁾	Detune offset in cents for carriers
114	Det offs Mod	-128.0...127.75 ⁽⁵⁾	Detune offset in cents for modulators
115	Det offs Op1	-128.0...127.75 ⁽⁵⁾	Detune offset in cents for operator 1
116	Det offs Op2	-128.0...127.75 ⁽⁵⁾	Detune offset in cents for operator 2
117	Det offs Op3	-128.0...127.75 ⁽⁵⁾	Detune offset in cents for operator 3
118	Det offs Op4	-128.0...127.75 ⁽⁵⁾	Detune offset in cents for operator 4
119	Det offs Op5	-128.0...127.75 ⁽⁵⁾	Detune offset in cents for operator 5
120	Det offs Op6	-128.0...127.75 ⁽⁵⁾	Detune offset in cents for operator 6
121	Det scal All	0.0...1.998 ⁽⁵⁾	Detune multiplier for all operators
122	Det scal Car	0.0...1.998 ⁽⁵⁾	Detune multiplier for carriers
123	Det scal Mod	0.0...1.998 ⁽⁵⁾	Detune multiplier for modulators
124	Det scal Op1	0.0...1.998 ⁽⁵⁾	Detune multiplier for operator 1
125	Det scal Op2	0.0...1.998 ⁽⁵⁾	Detune multiplier for operator 2
126	Det scal Op3	0.0...1.998 ⁽⁵⁾	Detune multiplier for operator 3
127	Det scal Op4	0.0...1.998 ⁽⁵⁾	Detune multiplier for operator 4
128	Det scal Op5	0.0...1.998 ⁽⁵⁾	Detune multiplier for operator 5
129	Det scal Op6	0.0...1.998 ⁽⁵⁾	Detune multiplier for operator 6
130	-		
131	-		
132	-		

133	-		
134	Waveform All	0...132 ⁽⁵⁾	Waveform select for all operators
135	Waveform Car	0...132 ⁽⁵⁾	Waveform select for carriers
136	Waveform Mod	0...132 ⁽⁵⁾	Waveform select for modulators
137	Waveform Op1	0...132 ⁽⁵⁾	Waveform select for operator 1
138	Waveform Op2	0...132 ⁽⁵⁾	Waveform select for operator 2
139	Waveform Op3	0...132 ⁽⁵⁾	Waveform select for operator 3
140	Waveform Op4	0...132 ⁽⁵⁾	Waveform select for operator 4
141	Waveform Op5	0...132 ⁽⁵⁾	Waveform select for operator 5
142	Waveform Op6	0...132 ⁽⁵⁾	Waveform select for operator 6

⁽¹⁾ : 0.0...127.5 - override NOTE ON velocity with the value (saturated to 127.0)
127.75 - use NOTE ON velocity

⁽²⁾ : semitones / notes

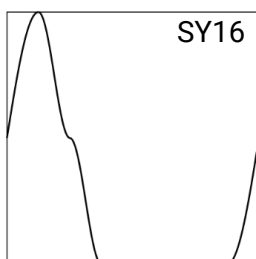
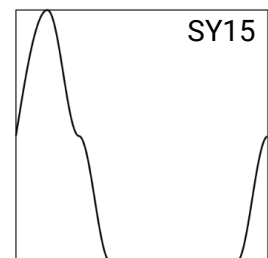
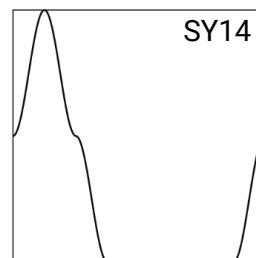
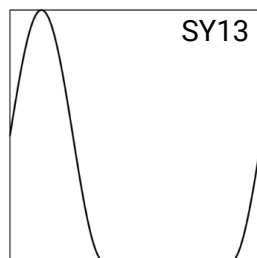
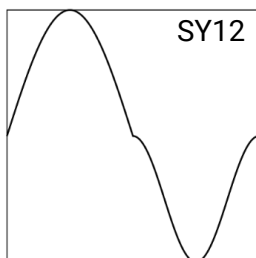
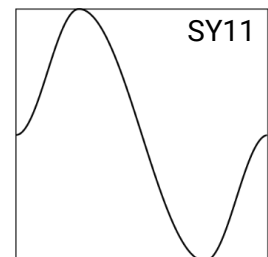
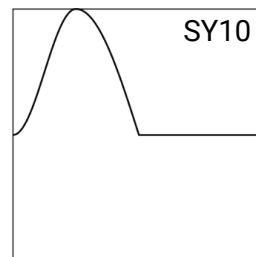
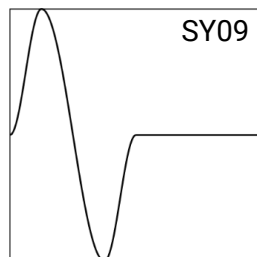
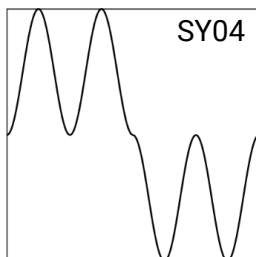
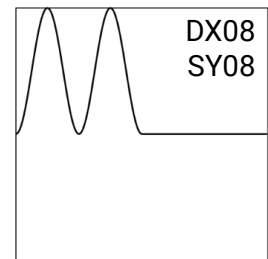
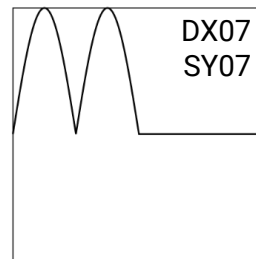
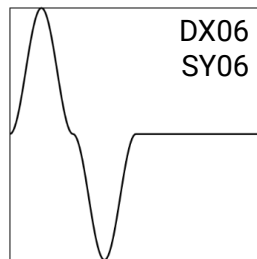
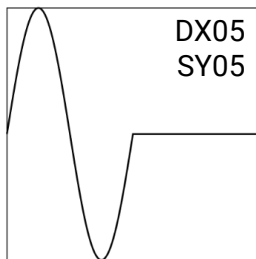
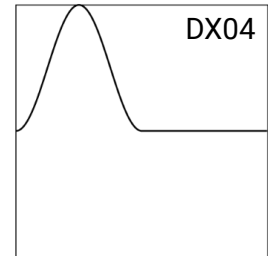
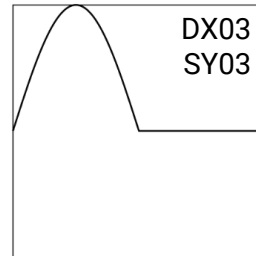
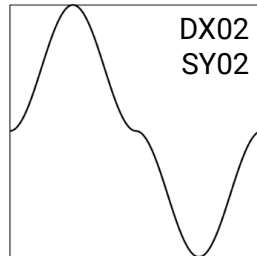
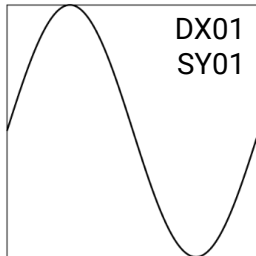
⁽³⁾ : Retain - keep voice feedback route
XtoY - feedback route source operator X to destination operator Y

⁽⁴⁾ : Retain - keep voice algorithm
dX01...dX32 - Yamaha DX series algorithms
oP33...oP40 - KORG opsix additional algorithms
SY01...SY45 - Yamaha SY77 / TG77 / SY99 algorithms

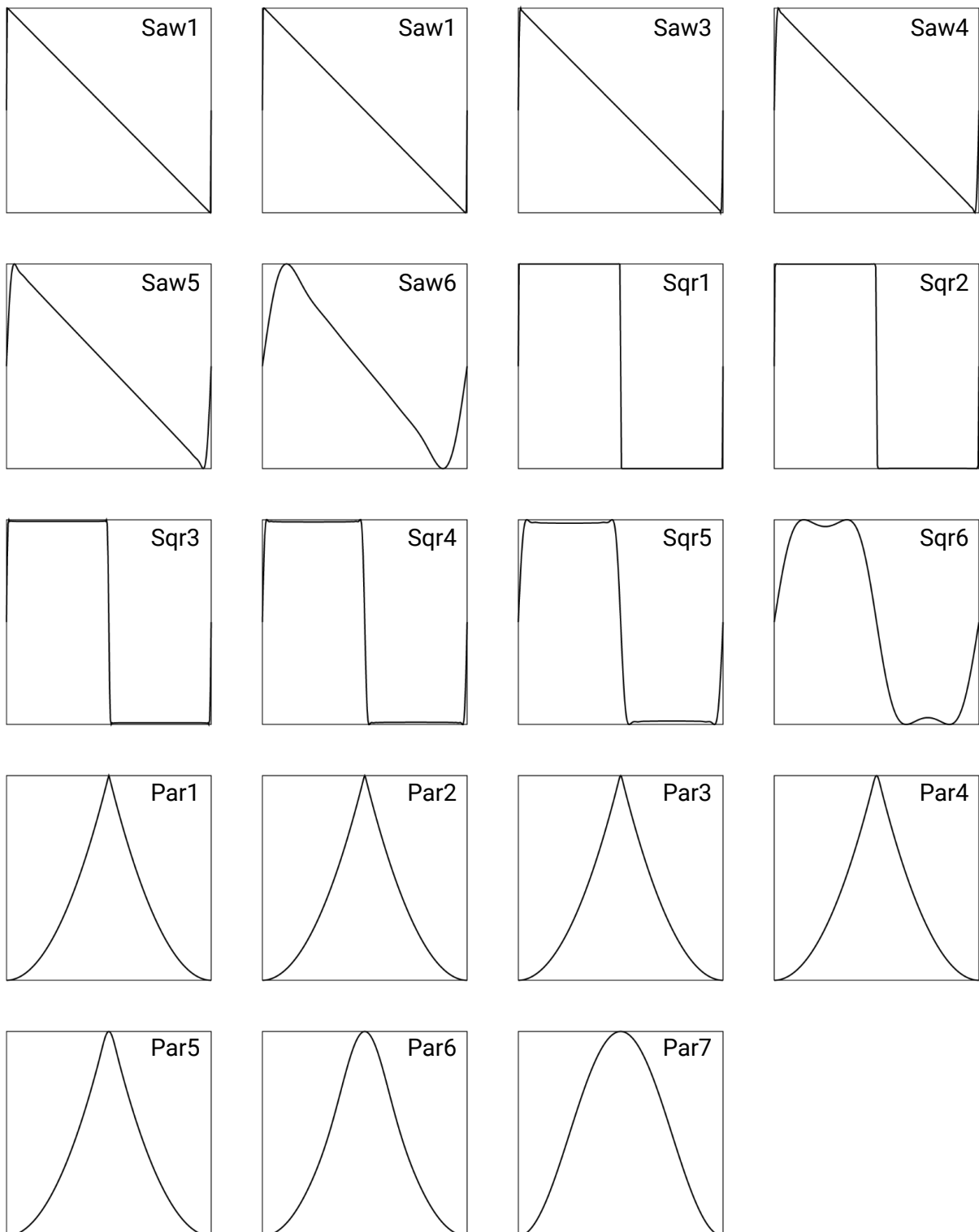
⁽⁵⁾ : effective in real time (otherwise on the next NOTE ON event)

Waveforms list

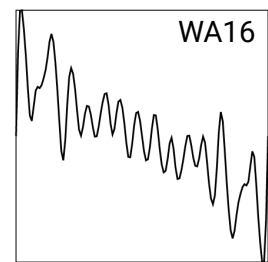
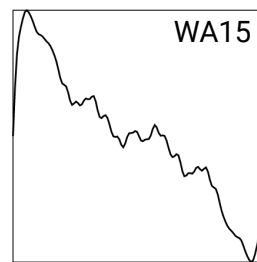
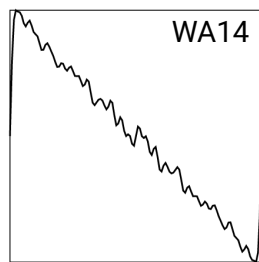
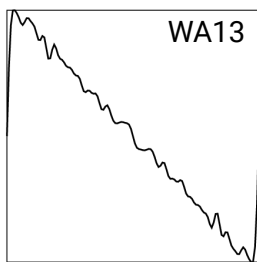
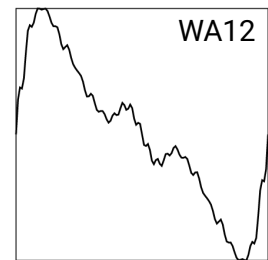
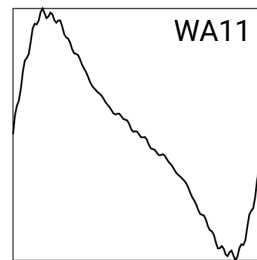
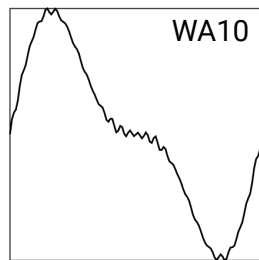
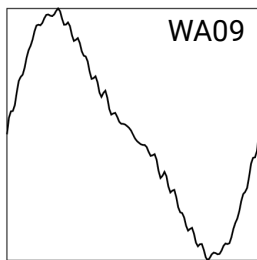
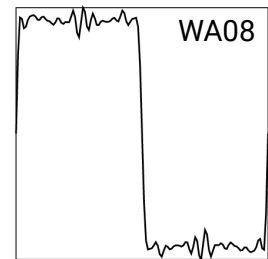
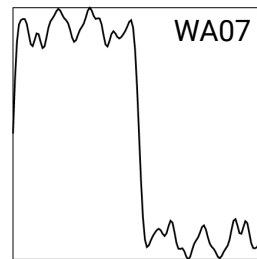
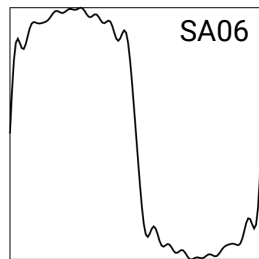
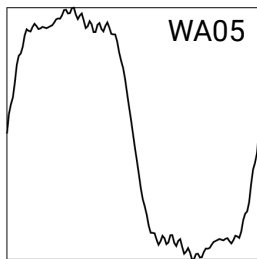
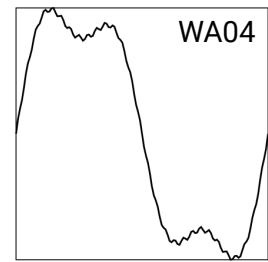
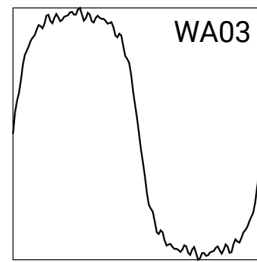
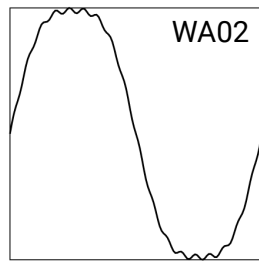
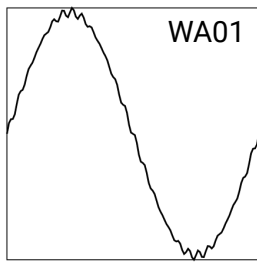
Yamaha DX11 / TX81Z and SY77 / TG77 / SY99 waves



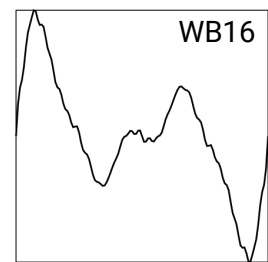
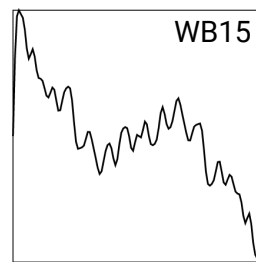
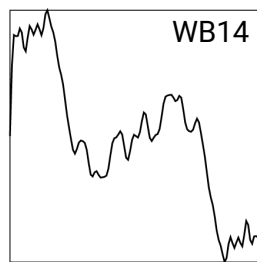
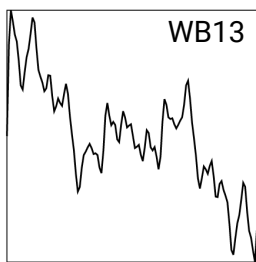
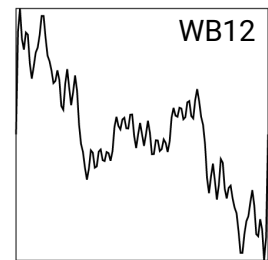
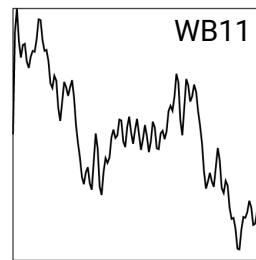
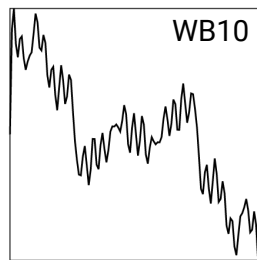
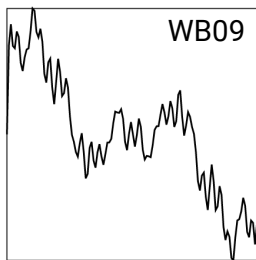
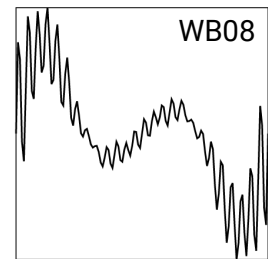
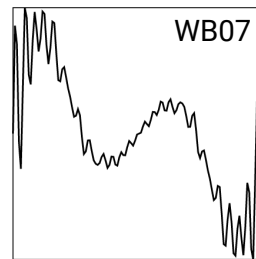
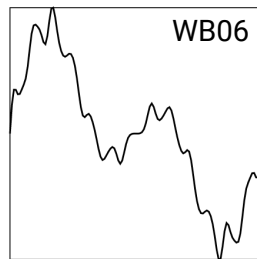
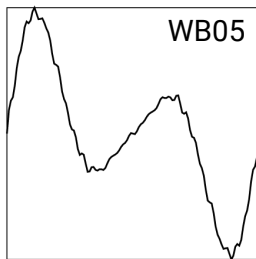
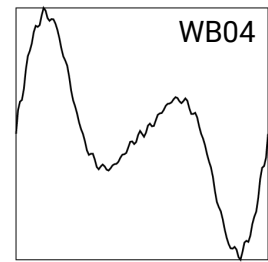
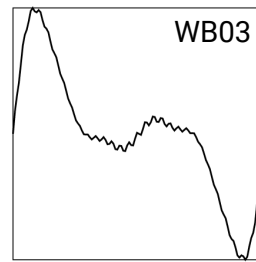
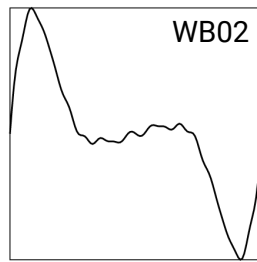
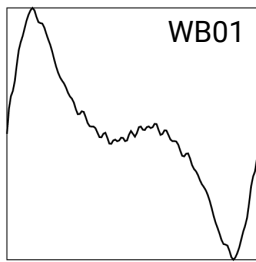
Sawtooth, Square and Parabolic waves



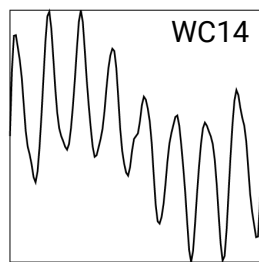
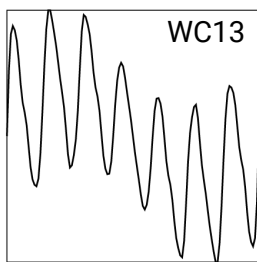
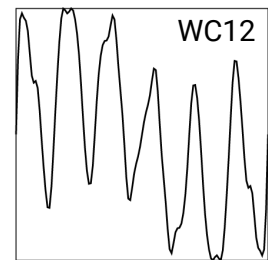
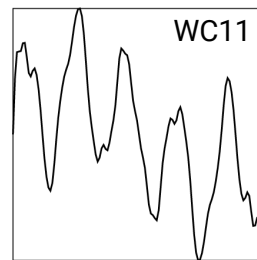
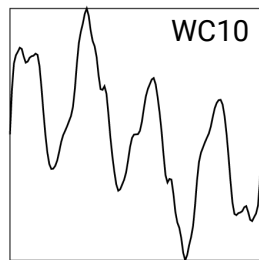
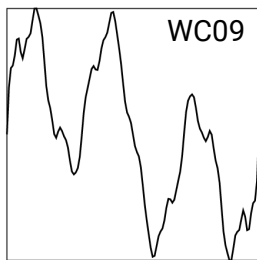
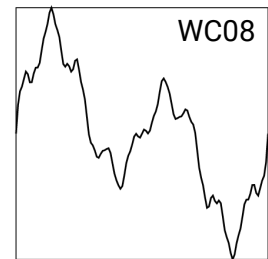
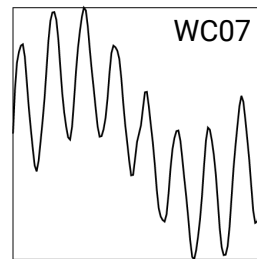
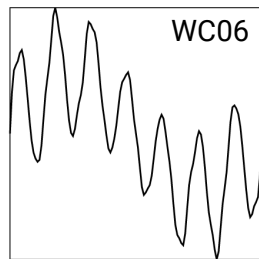
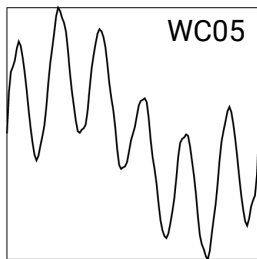
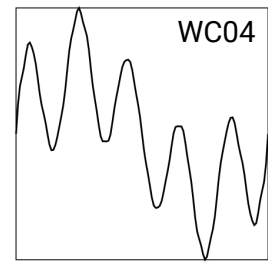
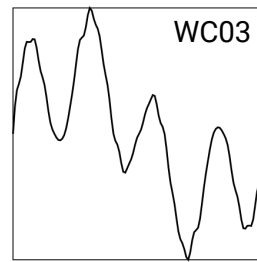
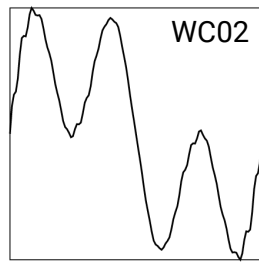
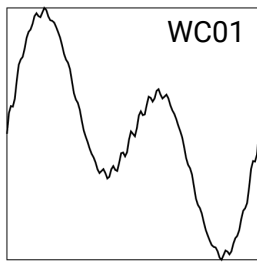
Waves bank A



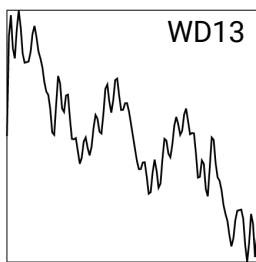
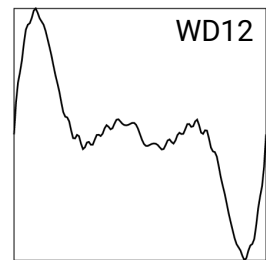
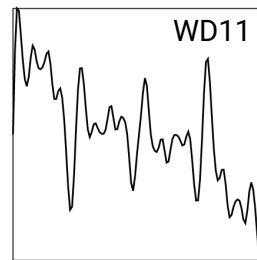
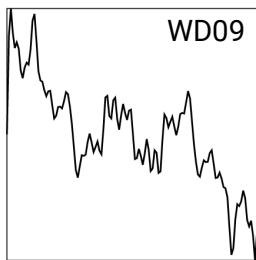
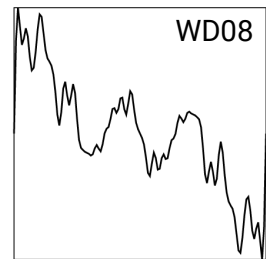
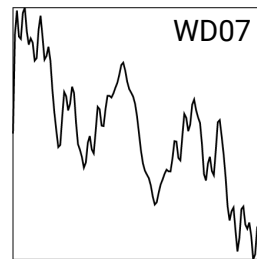
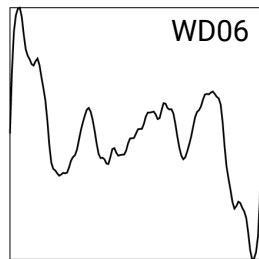
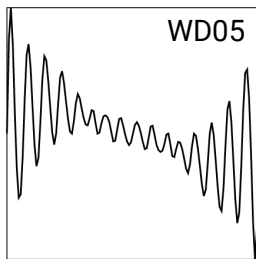
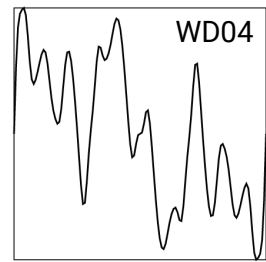
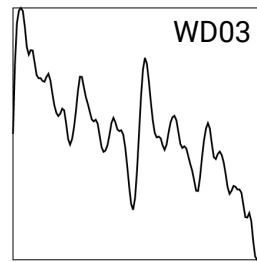
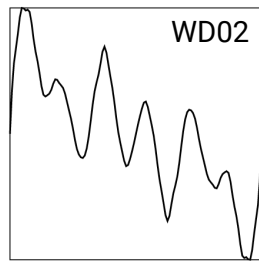
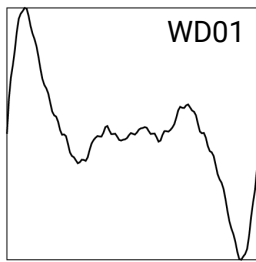
Waves bank B



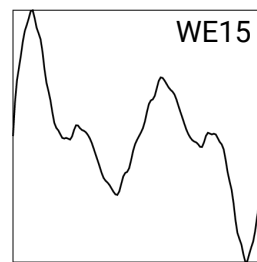
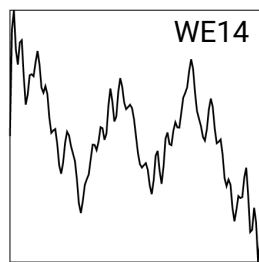
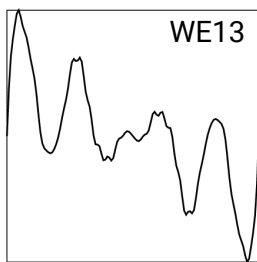
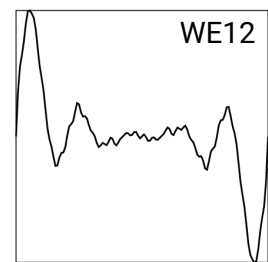
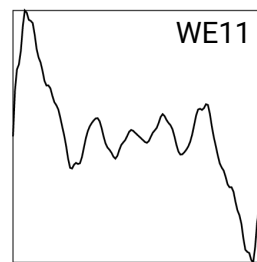
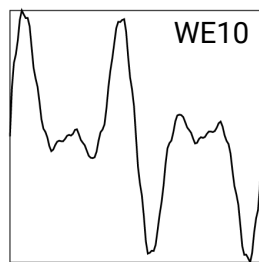
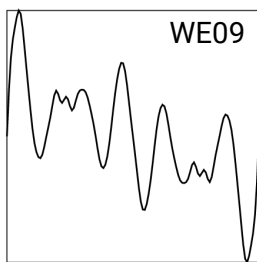
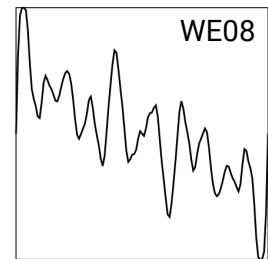
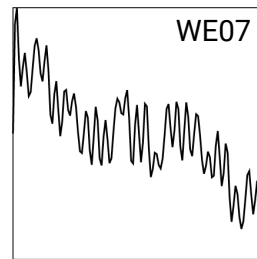
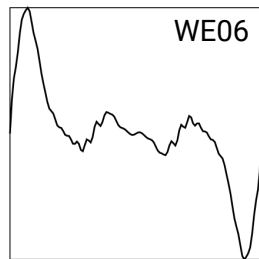
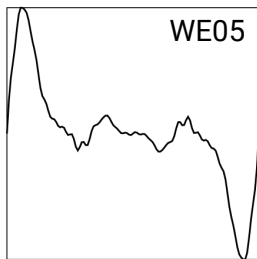
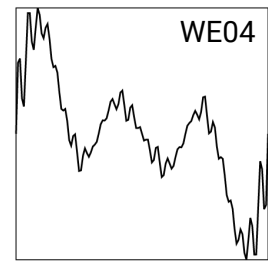
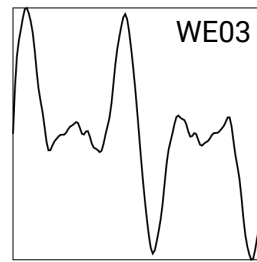
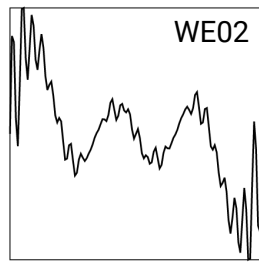
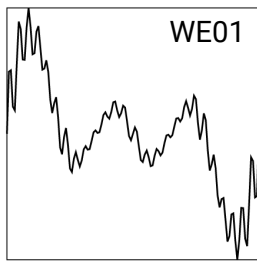
Waves bank C



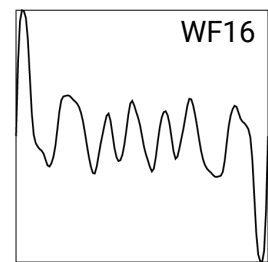
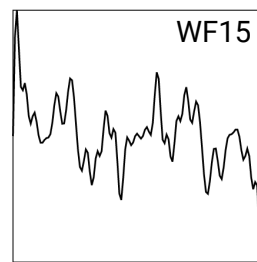
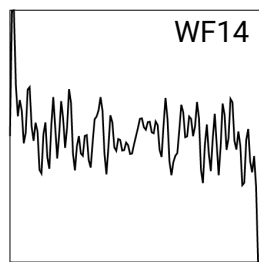
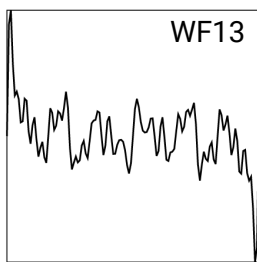
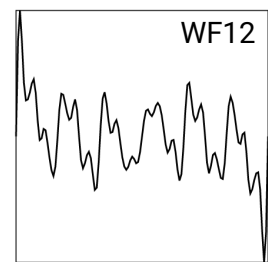
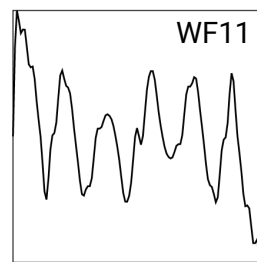
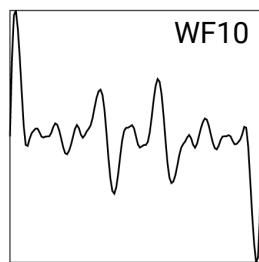
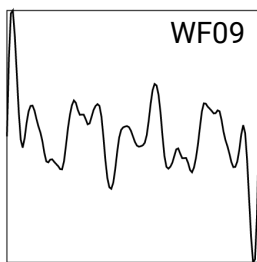
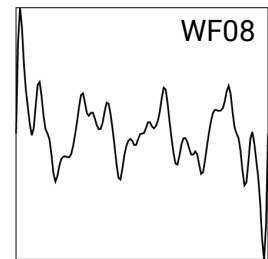
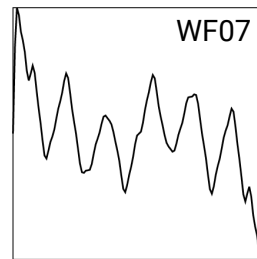
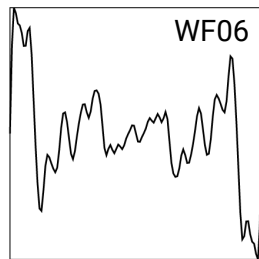
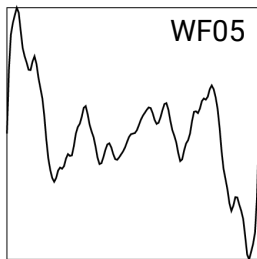
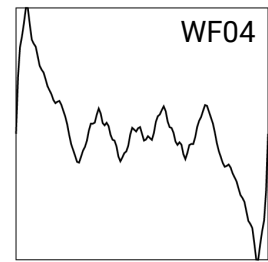
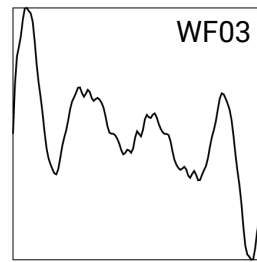
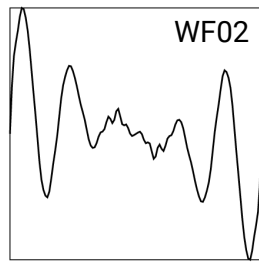
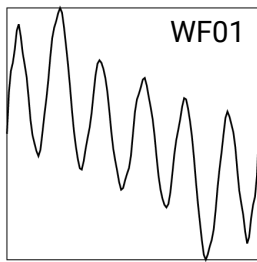
Waves bank D



Waves bank E



Waves bank F



Known issues and limitations

- Native Yamaha DX / TX series LFO, Amp and pitch modulations are not supported due to performance limitations.
- All ascending EG stages (e.x. typical Attack) are exponential. Implementing the reference semi-linear behavior will introduce computational complexity that is not currently affordable.
- NTS-1 mkII Shape/A and Alt/B parameters values resolution are reduced to 10-bit while saved/restored with KORG KONTROL Editor
- NTS-1 mkII CC/NRPN 14-bit values might not be mapped correctly to the corresponding parameter values, refer to <https://github.com/korginc/logue-sdk/issues/116>
- NTS-1 mkII fractional numbers might not be displayed correctly, refer to <https://github.com/korginc/logue-sdk/issues/118>